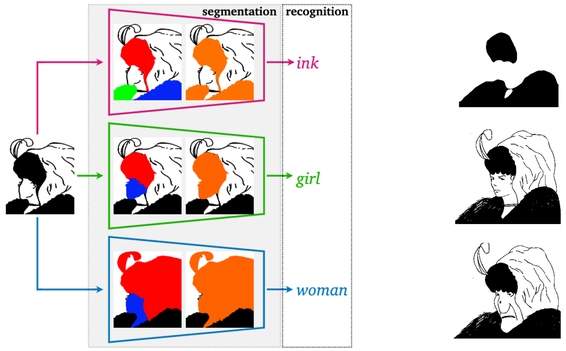
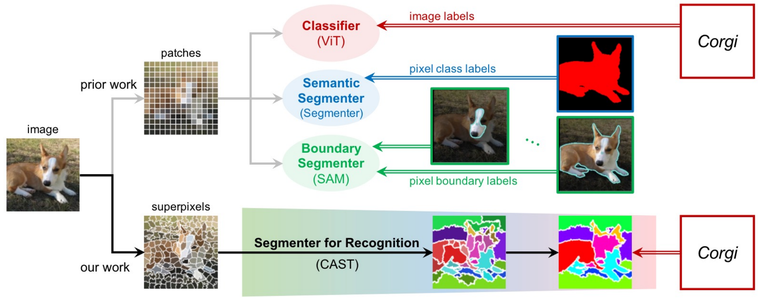
Learning Hierarchical Image Segmentation For Recognition and By Recognition

Tsung-Wei Ke * Sangwoo Mo * Stella X. Yu

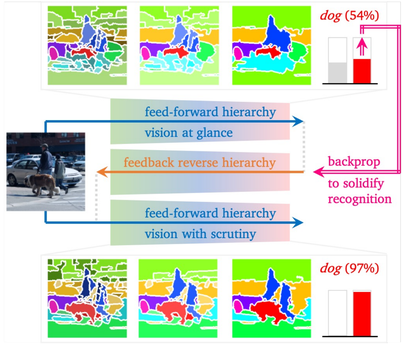
Consistency of Visual Parsing instead of Text Labels



Continuum of Internal Segmentation onto Recognition

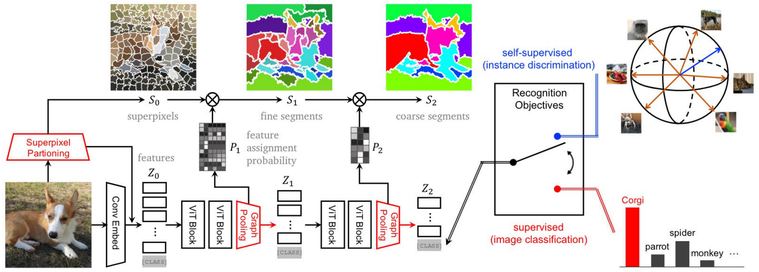


Concurrency of Segmentation and Recognition



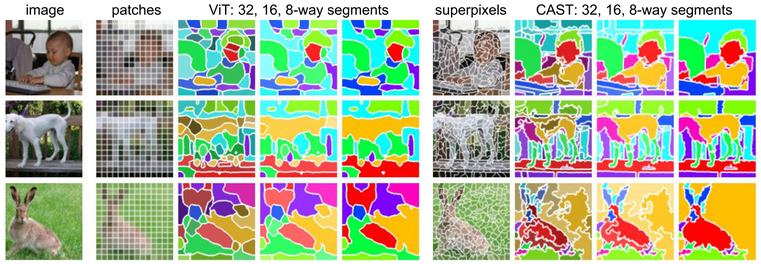
1. Always trained, tested, and adapted together
2. Segmentation substantiates recognition
3. Recognition leads segmentation

Our CAST = ViT w/ Superpixels + Graph Pooling

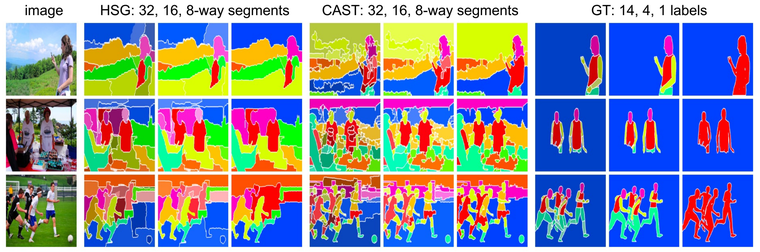


Segmentation as part **OF** **Recognition**
learned **BY** **Recognition**
uncovering part-whole **FOR** **Recognition**

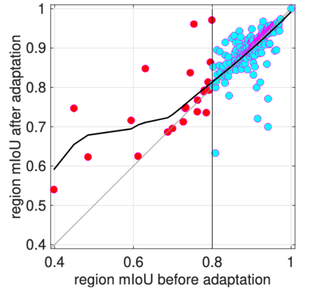
1. Unsupervised Part-Whole Discovery

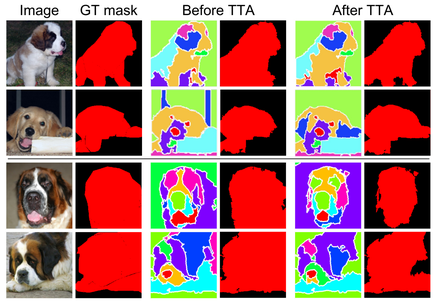




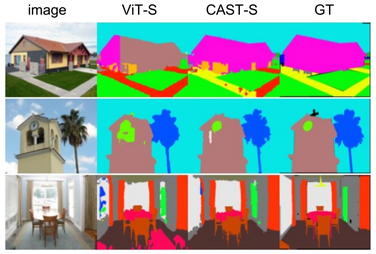


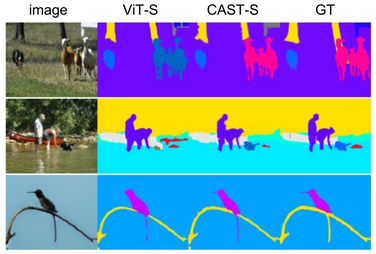
2. Solidify Recognition by Adapting Segmentation





3. Generalization and Efficiency





model	GFLOPS	region mIoU		boundary F-score	
SAM-B	677	18.03	10.15	20.71	7.25
SAM-H	3166	21.97	12.07	**32.66**	**11.82**
ViT-B	18	25.34	11.74	10.92	4.64
CAST-B	**13**	**29.66**	**13.20**	22.32	6.52

P R F	14 labels			4 labels			1 label		
HSG	20.7	18.6	19.6	24.1	30.6	26.9	20.5	36.1	26.2
CAST	**21.1**	**24.1**	**22.5**	**24.8**	**33.2**	**28.4**	**26.3**	**44.9**	**33.2**
gain	0.4	5.5	2.9	0.7	2.6	1.5	5.8	8.8	7.0

test on PASCAL-VOC	before tuning		after tuning	
ViT-S	30.9	16.1	65.8	40.7
ViT-S but with token pooling	34.5	19.8	67.2	41.9
ViT-S but with superpixels	32.2	21.2	66.5	46.7
CAST-S	**38.4**	**27.0**	**67.6**	**48.1**

Model	GFLOPS	IN-100	IN-1K
ViT-S	4.7	78.1	67.9
Swin-T	4.5	78.3	63.0
CAST-S	**3.4**	**79.9**	**68.1**

1. Beats ViT on unsupervised hierarchical segmentation on ImageNet
2. Beats SAM on unsupervised object segmentation on PartImageNet
3. Beats HSG on unsupervised human-body parsing on DensePose
4. Better downstream semantic segmentation
5. Better recognition and efficiency
6. Both superpixels and token pooling contribute to performance gains